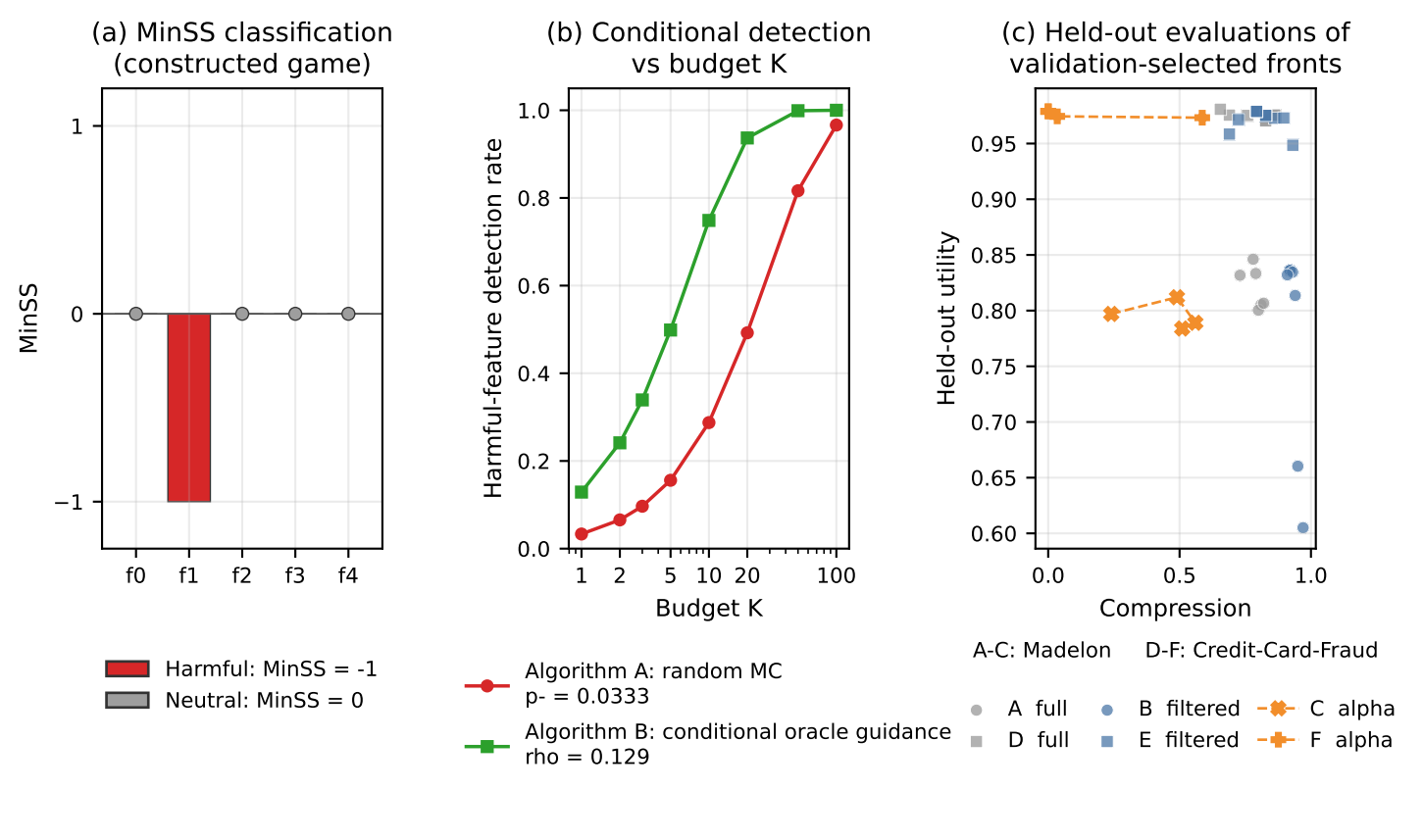
(a) MinSS classification
(constructed game)
MinSS
1
0
−1
f0
f1
f2
f3
f4
Harmful: MinSS = -1
Neutral: MinSS = 0
(b) Conditional detection
vs budget K
Harmful-feature detection rate
1.0
0.8
0.6
0.4
0.2
0.0
1
2
5
10
20
100
Budget K
Algorithm A: random MC
p- = 0.0333
Algorithm B: conditional oracle guidance
rho = 0.129
(c) Held-out evaluations of
validation-selected fronts
Held-out utility
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.0
0.5
1.0
Compression
A-C: Madelon
D-F: Credit-Card-Fraud
A full
B filtered
C alpha
D full
E filtered
F alpha